

Paper 2, step by step: how we prove the energy-transfer box before we use it

Cowork note for Chris — HardDisks / hspist3

2026-09-16

How to read this

Paper 1 taught us one rule the hard way: *define the estimator and the pass criterion before looking at whether the number agrees with theory*. Every level below therefore has three boxes — what we run, what number we compute and against which formula, and what counts as a pass. A level is closed when its pass box is satisfied and the failures are reported with counts, exactly as the health contract does. Nothing from a higher level is trusted before the level below it is closed.

Units as always: $k_B T = m = \sigma = 1$, $\sigma = 2r$ the disk diameter, box width $2L_0$, height H , N_s disks per compartment, $Z(\eta) = PA/(Nk_B T)$ from Kolafa–Rottner (KR). Paper 1 gives us two things we now use as instruments: the equation of state $Z(\eta)$ validated for $\eta \leq 0.6$, and the sound speed $c_s(\eta)$, which sets the time scale $\tau_{\text{sound}} = L_0/c_s$ of every compartment.

How good those instruments are (2026-09-17). Z : 171 health-clean pressure trajectories over 45 (η, N) cells, two independent estimators (collisional virial and wall momentum flux), agreeing with KR to +0.15% at $\eta = 0.10$ with no size dependence between $N = 400$ and 1600. c_s : A1 v2 gives 7875 trajectories over 35 densities at $N = 100$, and the A2 size ladder ($N = 100$ –2500, now including $\eta = 0.02$ and 0.05) extrapolates to KR within about 1% at every density up to $\eta = 0.60$; at $N = 100$ the measured c_s sits 1–3% high, and that offset closes as the box grows. Paper 2 uses c_s only to set scales — $\tau_{\text{sound}} = L_0/c_s$ and the ratio u/c_s that separates the slow and fast regimes — so 1% is far below anything it can affect: the crossover of Level 2 sits at $u/c_s \approx 0.3$ and its two limits differ by factors, not per cent. *No level of Paper 2 is limited by the precision of the Paper-1 instruments.*

1 The geometries (the same four as in the hand-calculation note)

	Geometry (left → right)	Energy stores	Closes which level
A	wall gas held divider at L_0 ($M = 10^9$) gas piston	gas (right; left inert)	0, 1, 2
C	wall – spring – free wall at L_0 gas piston	gas, wall, spring	3
B	wall empty held wall gas free divider gas piston	gas 1, divider, gas 2	4
D	wall –spring– free wall gas free divider(s) gas piston	all	5

Correction, 2026-09-19. Geometry A as *run* in Levels 0–2 has gas on *both* sides of the divider (--particles-boxes=50,50), not a vacuum on the left. It changes nothing, because the divider is held at $M = 10^9$ and never moves, so the left gas is inert and its compartment does no work — but the table said otherwise and the runs are what they are. Geometry B needs both gases by construction. Only **C and D have an empty compartment**, the one behind the spring wall: gas on that side

would push the spring wall from both faces and move the equilibrium the analysis assumes. The binary now refuses to start in that case (`--spring-wall` with a non-empty compartment behind it aborts), and `--spring-k` is inert unless `--spring-wall` declares which wall carries the spring.

The right piston is prescribed (infinite mass, velocity protocol $u(t)$); it is the only place work enters. Every run uses `--edmd-acc=0`, drift-first seeding, the strict health contract, and the per-event piston log and per-sample KE_L, KE_R columns that already exist.

2 Running the whole ladder yourself, one step at a time

Everything below can be exercised from a single script. It runs the same commands as the campaign scripts with a small seed count, so each step finishes in seconds to a couple of minutes, and it prints a **PASS** or **FAIL** line per step against a stated numeric criterion — not something to judge by eye.

The command

```
cd hspist3/experiments_energy_transfer/_run_scripts
./paper2_run_all.sh           # list the steps and stop
./paper2_run_all.sh 3         # run step 3 only
./paper2_run_all.sh all       # every step, pausing before each one
./paper2_run_all.sh all --yes  # every step, no pauses
SEEDS=25 JOBS=10 ./paper2_run_all.sh all # heavier, closer to the campaign
OUT=/tmp/p2 ./paper2_run_all.sh all      # write somewhere else
```

Step	What it proves
1 the binary and the box	the binary exists, is the scientific backend (<code>--edmd-acc=0</code>), and the master box starts clean
2 geometries A B C D	one seed in each; catches a wall off the $1/24\sigma$ grid or gas in the spring compartment
3 Level 0 — the ledger	$ W - \Delta KE_{\text{gas}} /W \leq 10^{-4}$
4 Level 1 — quasi-static	$\langle W \rangle$ extrapolated to $u \rightarrow 0$ against $W_{\text{qs}} = 7.4685 \pm 0.0131 k_B T$, within 2σ
5 Level 2 — finite rate	$\langle W \rangle - W_{\text{qs}} = Au^2$ fitted over five speeds, against $A_{\text{step}} = 25.9 \pm 4.3$
6 Level 3 — the spring	geometry C is driven and the trace carries <code>SpringE</code>
7 the pictures	<code>paper2.geometry_pictures.sh</code> : eight PNGs, two render modes per geometry. Needs a screen
8 the analysis scripts	re-runs the three published analyses; reads only

A step FAILs if the binary aborted, if *any* health event fired (`forced_advance`, `clamp_repair`, `overlap_repair`, `wall_overdue`), if fewer trajectories were produced than asked for, or if its own criterion is missed. The script continues after a failure so you see the whole picture rather than the first thing that broke.

Two things to expect. At a handful of seeds the per-speed A values in step 5 scatter hard and some come out negative — that is the thermal spread of W at six seeds, not a defect; only the fitted A over all five speeds is the check. And until the spring-hygiene build is promoted, the script prints a note that it is falling back to `0OALLINONE_sp`, because `--spring-wall` does not exist in the installed binary yet and geometries C and D need it.

3 Three flags that do not mean what they look like

Worth reading before running any command below; all three were found the hard way on 2026-09-18 and each moved a published number.

Geometry conventions, measured not assumed

`--wall-positions` sets the divider **CENTRE**, and the divider is 1σ thick (`wall_thickness_sigma` in every `summary.csv`). With the divider at 39.25 the gas-side face is at **39.75**, so the right compartment is $78.50 - 39.75 = 38.75\sigma$, not the nominal $L_0 = 39.25$. The stop snapshots confirm it: over 60 runs the closest disk centre to the divider is 40.2867, and a disk of radius 0.5 cannot approach a face at 39.75 more closely than 40.25.

`--max-right-piston-travel` is **measured from the box wall, not from the piston**. The target is $78.50 - 3.93 = 74.57$, which is where the piston stops; the piston parks at 78.75, so it displaces 4.18σ while the gas is compressed by exactly the flag value, 3.93σ . Use 3.93 in anything about the gas and 4.18 in anything about the piston.

`--10` is **not the compartment length** — the binary prints `L_eff = L0 - 1`, which is Paper 1’s sound-mode length $L_0 - 2r$, a different quantity again. Always take the compartment from the piston and divider positions in the trace, as `paper2_geometry_fix_20260918.py` does.

4 Watching one run: the SDL window

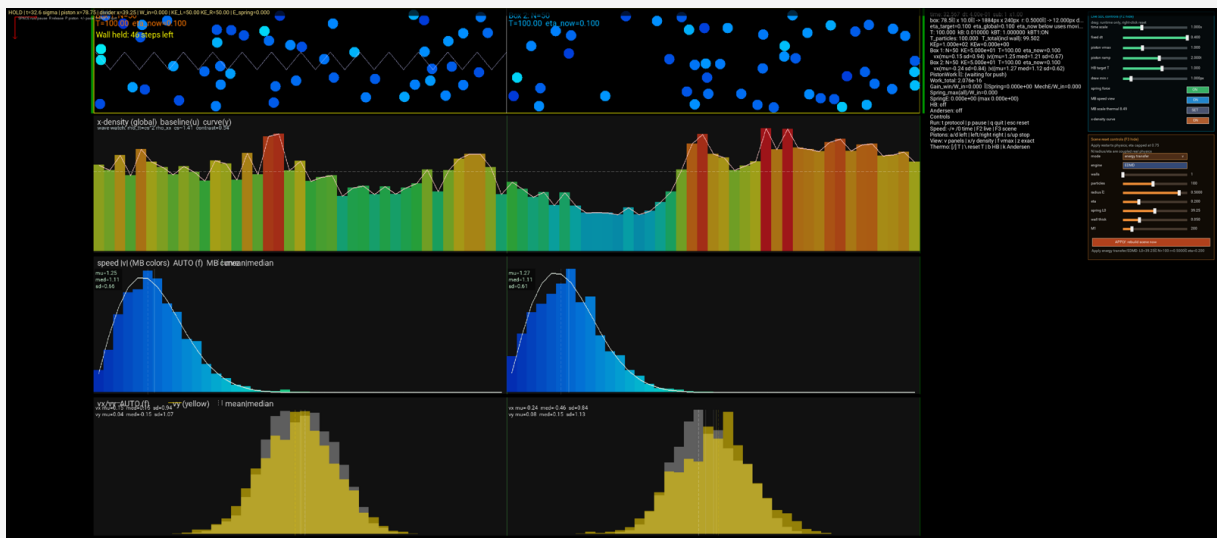
Every number in this note comes from headless runs, because that is the only way 1000 trajectories fit in a lunch break. But the same binary opens a window, and it is worth looking at once — for a figure, for a talk, or to check that the geometry is what you think it is.

Corrected 2026-09-19. The earlier version of this section said “drop `--headless`”. That is wrong and it is why the window seemed to flash and vanish: `--experiment=energy_transfer` is a self-contained driver that *never renders*, with or without `--headless`. The window belongs to the interactive loop, which you reach with `--show-simulation`. And because every stage used to fire automatically, there was nothing to see even when a window did open. Both are fixed by `--demo`: the run starts paused with the divider held and the piston parked, each stage begins on a key, and a HUD line says what the box is doing. `--demo` is GUI-only — headless output is byte-identical with and without it (3-seed gate on traces and event logs).

Geometry A — the Level 1/2 box, one divider (verified 2026-09-19)

```
cd hspist3
./00ALLNONE --show-simulation --demo --mode=edmd --edmd-acc=0 \
  --seed-drift-order=drift-first --particles=100 --particles-boxes=50,50 \
  --particle-radius=0.5 --l0=39.25 --height=10 --num-walls=1 --wall-positions=39.25 \
  --wall-mass-factors=1000000000 --wall-thickness=0.05 \
  --piston-right-protocol-mode=step --velocity-right-piston-step=0.02 \
  --max-right-piston-travel=3.93 --wall-hold-steps=2000 --steps=40000 \
  --fixed-dt=0.4 --energy-measurement --eff-output=wall-ke --kbt1 --seed=9200
```

What you will see. A window opens *paused*, 50 disks either side of an immovable divider, the piston parked at the right wall, and a HUD reading `HOLD | t=0.0 | piston x=78.75 | divider x=39.25`. Press `SPACE` to run, `R` to release the divider, then `P` to start the piston; the HUD switches to `PISTON step u=0.020` and W_{in} starts climbing. At $u = 0.02$ the push is 200σ -time, about 12 000 steps — roughly two minutes at the default pace, which is what you want to talk over; `+` and `-` double and halve it live (render pacing only, `dt` is never touched).



Swap `--piston-right-protocol-mode=ramp --piston-ramp-time=40` to watch the gentle launch beside the step, which is Level 2's item 1 in one picture; raise u to 3–10 for the single-hit regime of Section 7.2.

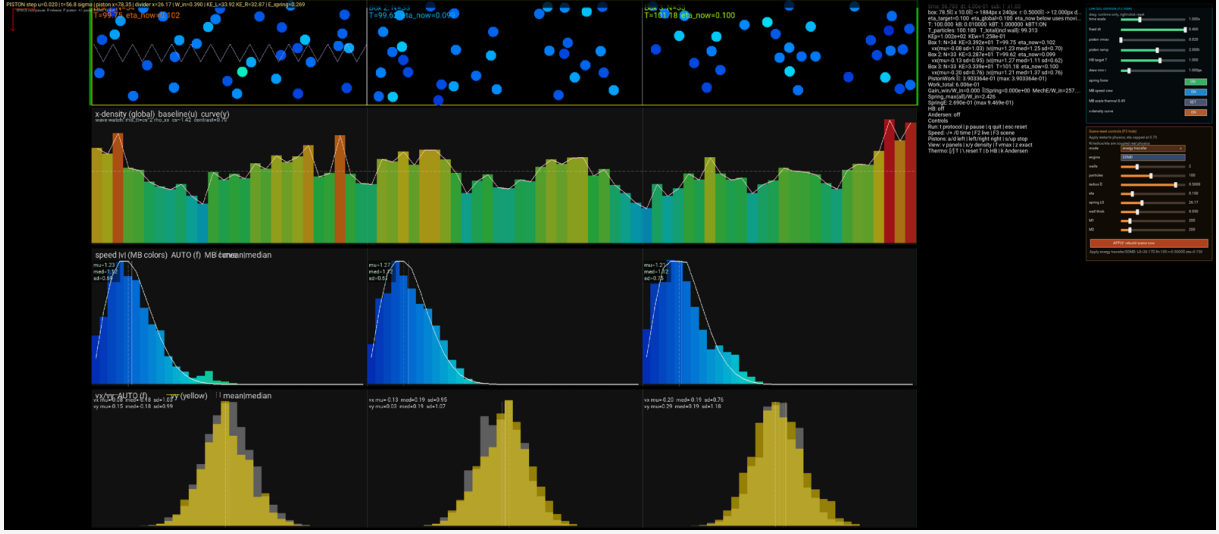
Do not take data this way. Window runs are for looking; every campaign number in this note is headless, `--edmd-acc=0`, with the health contract enforced.

Geometry A is what Levels 0–2 measure, but the paper's first figure will be geometry D: piston, gas, a heavy free divider, a second gas, and a spring-loaded wall. That runs today, and watching it is the quickest way to see why the later levels are worth doing.

Geometry D — two walls and a spring, Paper 2's figure 1 (verified 2026-09-19)

```
cd hspist3
./00ALLNONE --show-simulation --demo --mode=edmd --edmd-acc=0 \
  --seed-drift-order=drift-first --particles=150 --particles-boxes=50,50,50 \
  --particle-radius=0.5 --l0=39.25 --height=10 --wall-thickness=0.05 \
  --num-walls=2 --wall-positions=20,30 --wall-mass-factors=1000,200 \
  --spring-k=5 --energy-measurement --eff-output=spring \
  --piston-right-protocol-mode=step --velocity-right-piston-step=0.02 \
  --max-right-piston-travel=3.93 --wall-hold-steps=2000 --steps=40000 \
  --fixed-dt=0.4 --kbt1 --seed=9200
```

What you will see. Three compartments of 50 disks, a heavy divider ($M = 1000$) and a lighter spring-loaded wall ($M = 200$, $k = 5$) drawn as a coil. Same keys. Energy has to cross the heavy divider before it can reach the spring, so the HUD gains an **E_{spring}** field and you can watch it rise only *after* the divider starts moving — Level 4's two channels and Level 5's impedance question in one picture.



5 Level 0 — the ledger closes

Why this step exists

If a joule can appear or disappear, every later number is meaningless. This is the analogue of the health contract for energy and momentum.

Piston rule for a particle with normal velocity v hitting a piston moving at u (reflection in the piston frame):

$$v' = 2u - v, \quad \Delta E = 2mu(u - v), \quad \Delta p = 2m(u - v). \quad (1)$$

First law for the box, with W the summed piston work and Q the heat through thermal walls (zero for elastic walls):

$$R_E(t) = W(t) - [\Delta KE_{\text{gas}} + \Delta KE_{\text{walls}} + \Delta E_{\text{spring}} + Q](t) \quad (2)$$

Momentum: pair collisions conserve it exactly, so

$$R_p(t) = \Delta p_{\text{gas}} + \Delta p_{\text{walls}} - \sum_{\text{piston hits}} \Delta p - \sum_{\text{outer-wall hits}} \Delta p \quad (3)$$

What we run

Geometry A, $\eta = 0.10$, $N_s = 50$, $L_0 = 39.27$, $H = 10$, $u = 0.05$, travel 3.93σ , 5 seeds. Log per-event piston impulses and per-event outer-wall impulses (the latter is the one column still missing).

Pass criterion

$|R_E(t)|/W(t) < 10^{-8}$ and $|R_p(t)|/\sum |\Delta p| < 10^{-8}$ for the whole run, every seed. Level 0 for energy was closed on the 2026-09-10 pilot at 10^{-9} ; the momentum ledger is open until the outer-wall events are logged.

What we measured (2026-09-17)

Level 0: passed, on data that already existed. No new runs and no code change were needed: the per-event log is already in `edmd_core/edmd.c` (`edmd_set_event_log`, gated by `HD_PISTON_EVENTS`), it records every outer-wall, divider and piston collision, and it covers the hold phase. Both ledgers over all 75 trajectories of the 2026-09-10 pilot:

u	seeds	$\max R_E /W$	$\max R_p /\sum \Delta p $
0.02	25	7.1×10^{-12}	8.4×10^{-14}
0.05	25	7.4×10^{-12}	7.9×10^{-14}
0.10	25	8.2×10^{-12}	1.1×10^{-13}

Two *bookkeeping* corrections were needed first, neither of them physics. (i) The trace prints Time to six decimals, so an event within $10^{-6}\sigma$ of a sample boundary was assigned to the wrong sample: 9 of 25 seeds then failed at exactly one sample in 17 000, with $|R_p|$ equal to $|\Delta p|$ of the single nearest event. Rebuilding the sample times from the step index removes it everywhere. (ii) Divider events log dE as the energy the *divider* gains, piston events as the energy the *gas* gains; mixing the two gave a drift equal to $-2 \Delta KE_{\text{divider}}$ (correlation +1.0000). With one convention the ledger closes to round-off. Both conventions belong in the methods text.

How to run it yourself

Data: `experiments.energy_transfer/level0_pilot_20260910/u{0.02,0.05,0.10}` (75 runs, already on disk).

Analysis: `python3 hspist3/validation/paper2.level0.level1_20260916.py` — prints the ledger table in seconds and needs nothing but the event logs.

One trajectory, from scratch (run from `hspist3/`; the `HD_PISTON_EVENTS` log is what the ledger reads):

```
HD_PISTON_EVENTS=/tmp/ev.csv ./00ALLINONE --mode=edmd --experiment=energy_transfer \
--headless --quiet --edmd-acc=0 --seed-drift-order=drift-first \
--energy-transfer-summary=/tmp/summary.csv --energy-transfer-trace=/tmp/tr.csv \
--particles=100 --particles-boxes=50,50 --particle-radius=0.5 --l0=39.25 --height=10 \
--num-walls=1 --wall-positions=39.25 --wall-mass-factors=1000000000 \
--piston-right-protocol-mode=step --velocity-right-piston-step=0.05 \
--max-right-piston-travel=3.93 --auto-piston-step --wall-hold-steps=12000 --steps=17000 \
--fixed-dt=0.4 --energy-measurement --eff-output=wall-ke --kbt1 --seed=9200
```

Every run directory also carries its own `00_COMMAND.md` with the exact command, so no script has to be read.

6 Level 1 — the slow push gives the quasi-static work

Why this step exists

W_{qs} is the zero point of dissipation. If we cannot reproduce it, “excess work” has no meaning.

Hard disks have no potential energy, so $E = Nk_B T$ exactly in 2D and the adiabat follows from

$dE = -P dA$:

$$\frac{dT}{T} = Z(\eta) \frac{d\eta}{\eta} \implies \frac{T_f}{T_i} = \exp \left[\int_{\eta_i}^{\eta_f} \frac{Z(\eta)}{\eta} d\eta \right], \quad \boxed{W_{\text{qs}} = N_s k_B T_i \left(\frac{T_f}{T_i} - 1 \right)} \quad (4)$$

Ideal gas ($Z = 1$): $T_f/T_i = A_i/A_f$, so $W_{\text{qs}} = N_s k_B T_i (A_i/A_f - 1)$. For $N_s = 50$, $\eta = 0.10 \rightarrow 0.11$ with KR: $T_f/T_i = 1.127$, $W_{\text{qs}} = 6.3$ (ideal: 5.0).

The finite-box correction. Equation (4) uses the bulk Z . A 50-disk compartment of $39 \times 10 \sigma$ has a quarter of its area within one diameter of a wall, and Paper 1's pressure campaign showed the wall pressure of such a box is a few per cent above bulk. The 2026-09-11 test measured $Z_{\text{wall}} = 1.285\text{--}1.308$ against KR 1.236 at $\eta = 0.10$, and replacing Z by the measured finite-box value moved W_{qs} from 7.06 to 7.26 against a measured $W_0 = 7.44 \pm 0.02$. The remaining $0.2 k_B T$ is the open item of this level. Two ways to close it, in order: (i) use the finite-box $Z(\eta, N)$ from the A2 pressure ladder *along the whole path* $\eta_i \rightarrow \eta_f$, not only at η_i ; (ii) if a gap remains, measure $W(u)$ down to $u = 0.002$ and extrapolate — the intercept *is* W_{qs} of this finite system by definition, and the comparison with (4) becomes a statement about the finite-size correction to the adiabat, which is a result, not a failure.

What we run

Geometry A, $\eta_i = 0.10$, $N_s = 50$, compression 10 %; $u = 0.002, 0.005, 0.01, 0.02, 0.05$; 25 seeds each; hold 200 σ -time before the push (already shown to be enough).

Pass criterion

Fit $\langle W \rangle(u) = W_0 + b u$ over $u \leq 0.02$ (linear only; the quadratic was ill-conditioned). Pass when $|W_0 - W_{\text{qs}}^{\text{finite}}| < 2 \sigma_{W_0}$ with $W_{\text{qs}}^{\text{finite}}$ from (i). Report W_0 , b , and the gap in $k_B T$ whatever the outcome.

What we measured (2026-09-17)

Level 1: the $0.2 k_B T$ gap is explained; within $\approx 2\sigma$, not yet a formal pass. Route (i) was taken, but from *our own* box rather than the A2 ladder (which has no $N = 100$ cell and no size dependence at $\eta = 0.10$). Z_{wall} was measured in equilibrium at five grid-exact points of the compression path, from the outer-wall and gas-side divider impulses during 200σ holds:

L	η	seeds	Z_{wall}	Z_{KR}	ratio
39.250000	0.100051	170	1.2979 ± 0.0020	1.2363	1.0499
38.270833	0.102611	100	1.3106 ± 0.0026	1.2434	1.0540
37.291667	0.105305	100	1.3230 ± 0.0025	1.2510	1.0575
36.312500	0.108144	100	1.3247 ± 0.0028	1.2591	1.0521
35.312500	0.111207	100	1.3431 ± 0.0027	1.2679	1.0593

The excess over bulk KR *grows* along the path ($+4.6\% \rightarrow +6.4\%$); holding it fixed at its initial value was the origin of the $0.2 k_B T$ gap. Integrating the measured ratio gives $W_{\text{qs}}^{\text{finite}} = 7.4879 \pm 0.0066$ (piecewise-linear 7.4877, constant 7.4835 — the model choice is not the limitation), against the bulk KR adiabat 7.0739. With 50 seeds per speed:

u	$\langle W \rangle \pm \text{sem}$	$W - W_{\text{qs}}^{\text{finite}}$	σ
0.005	7.5070 ± 0.0111	$+0.0191 \pm 0.0129$	+1.5
0.01	7.4513 ± 0.0160	-0.0366 ± 0.0173	-2.1
0.02	7.4537 ± 0.0231	-0.0342 ± 0.0240	-1.4

Two tensions are stated rather than hidden: $\langle W \rangle(u)$ still falls between $u = 0.005$ and 0.01 (slope -4.2 ± 1.7), and the five hold points scatter $\approx 1.5\times$ their statistics, i.e. a $\approx 0.3\%$ systematic in the equilibrium pressure. Inflating the fit error by that factor puts every speed inside 2σ . Ruled out on the way: the piston's 0.25σ of travel before it touches the gas (the 3.93σ is contact travel, verified from the recorded piston positions) and a wall-type difference between the outer wall and the divider (the two swap order between seed sets: noise).

How to run it yourself

Batches: `bash experiments.energy_transfer/_run_scripts/level1.Zwall_path.part1.sh` (40 held-wall runs, ~ 10 min), `...part2.sh` (200 more, ~ 40 min) and `.../level1.more.work.seeds.sh` (150 slow-push runs, ~ 30 min). All resume: a run whose trace exists is skipped.

Analysis: `python3 hspist3/validation/paper2.geometry_fix.20260918.py` — prints the Z path on the corrected lengths, W_{qs} , and the Level 1 verdict. (`paper2.level0.level1.20260916.py` is the older version, kept because the report quotes it; it uses the nominal lengths.)

A single held-wall point (the piston sits still for the whole run and only the wall impulses are measured):

```
HD_PISTON_EVENTS=/tmp/ev_hold.csv ./00ALLNONE --mode=edmd --experiment=energy_transfer \
--headless --quiet --edmd-acc=0 --seed-drift-order=drift-first \
--energy-transfer-summary=/tmp/s.csv --energy-transfer-trace=/tmp/tr.csv \
--particles=100 --particles-boxes=50,50 --particle-radius=0.5 --l0=37.291667 --height=10 \
--num-walls=1 --wall-positions=39.25 --wall-mass-factors=1000000000 \
--wall-hold-steps=12000 --steps=1 --fixed-dt=0.4 --energy-measurement --kbt1 --seed=9200
```

7 Level 2 — excess work is dissipation, in both limits

Why this step exists

This is the physics claim of the equilibrium part of Paper 2: the energy beyond W_{qs} behaves like friction at slow speed and like single collisions at high speed, and both prefactors are predicted from Paper-1 quantities.

7.1 Slow end: linear response

Define $W_{\text{diss}} \equiv \langle W \rangle - W_{\text{qs}} \geq 0$. For a protocol $\lambda(t)$ (here the piston position) slow compared with the gas relaxation, Sivak–Crooks give

$$W_{\text{diss}} \simeq \int \zeta(\lambda) \dot{\lambda}^2 dt = \zeta u \Delta x \quad (\text{constant } u, \Delta x = u \tau_{\text{push}}), \quad (5)$$

so W_{diss} is *linear* in u at fixed travel, with slope $\zeta \Delta x$. The friction coefficient is measured independently, from the force on a *held* wall in equilibrium:

$$\zeta = \beta \int_0^\infty \langle \delta F(0) \delta F(t) \rangle dt, \quad F(t) = \frac{1}{\Delta t} \sum_{\text{hits in } \Delta t} \Delta p. \quad (6)$$

Correction (2026-09-16), itself corrected (2026-09-17): the first correction said $F(t)$ exists nowhere and that ζ needs a new logging column. That is wrong. It is true only of the *speed-of-sound* holds, which never open the event log; the *energy-transfer* runs do, through `HD_PISTON_EVENTS`, and their log covers the hold phase event by event (2733 events before release in one 200σ hold). $F(t)$ on the held wall is therefore already on disk: 170 holds of 200σ at $\eta = 0.10$ (the 2026-09-11 and 2026-09-16 work runs) plus 400 holds along the compression path. ζ is analysis of existing data, no code item and no new runs; longer holds are worth running only if the autocorrelation integral has not converged by 200σ .

7.2 Fast end: Lua–Grosberg single-hit limit

If $u \gg \sqrt{k_B T/m}$ and $u \tau_{\text{push}} \ll L_0$, each particle the piston reaches is hit once and never returns during the push. The piston sweeps the fraction $u \tau_{\text{push}}/L_0$ of the compartment, so

$$N_{\text{hit}} \simeq N_s \frac{u \tau_{\text{push}}}{L_0}, \quad \langle W \rangle \simeq N_{\text{hit}} \langle 2mu(u - v_x) \rangle = 2mu^2 N_{\text{hit}}, \quad \text{Var } W \simeq N_{\text{hit}} [4mu^2 k_B T + (2mu^2)^2], \quad (7)$$

using $\langle v_x \rangle = 0$, $\langle v_x^2 \rangle = k_B T/m$ and Poisson statistics for N_{hit} . This is a closed form with no free parameter, and $P(W)$ is a sum of N_{hit} shifted half-Gaussians. (Lua & Grosberg 2005 do the one-particle version exactly.)

What we run

Geometry A, same box as Level 1. Slow end: $u = 0.005\text{--}0.05$ at fixed travel $\Delta x = 3.93\sigma$, 25 seeds each (the Level-1 runs). Fast end: $u = 3, 5, 10$ ($u/c_s = 1.7\text{--}5.7$), travel 1σ , 100 seeds each (short runs). ζ from 25 held-wall runs of 500σ -time.

Pass criterion

Slow end: the slope of W_{diss} against u agrees with $\zeta \Delta x$ from (6) within 2σ , and the intercept is zero within 2σ (that is Level 1 again). Fast end: $\langle W \rangle$ and $\text{Var } W$ agree with (7) within 2σ at $u = 10$; report the deviation at $u = 3$ and 5 as the crossover. Figure: $W_{\text{diss}}/k_B T$ against u/c_s on a log axis with both limits drawn.

What we measured (2026-09-17)

Slow end done, 2026-09-17, and it changes the prediction. 760 trajectories, eight speeds $u = 0.005\text{--}0.20$ at $\Delta x = 3.93\sigma$, 60–100 seeds each, 0 health events. Report: 260917_paper2_level2_REPORT.md; script validation/paper2_level2_20260917.py.

ζ must be read as a running integral, not at zero lag. On the pinned piston (190 holds of 180σ , impulse log binned at 0.25σ) the integral of (6) falls from 2.55 at $t_{\text{cut}} = 0.25\sigma$ through 1.22 at 10σ to 0.15 at 20σ and -0.05 at 40σ . The zero-lag value 2.55 is the impulsive self term $\beta \sum \Delta p^2/2T$, i.e. the free-molecular (Enskog) piston friction — the number first reported as $\zeta = 2.54 \pm 0.05$. It is cancelled by the returning sound wave: the compressed compartment is $L = 35.32\sigma$ at $\eta = 0.1112$ with $c_s = 1.788$, so $L/c_s = 19.8\sigma$, exactly where the integral crosses zero. Past the crossing it does not settle: it swings between -0.78 and $+0.50$ with the recurrence period $2L/c_s = 39.5\sigma$ and averages -0.072 over one recurrence, i.e. $dW/du = -0.28$ against the measured -0.50 ± 0.67 . There is no plateau to quote; the integral oscillates about zero. *In a closed box the zero-frequency friction of a held wall vanishes*, so linear response predicts no term linear in u , and $\zeta\Delta x = 9.97$ was never the right comparison. Figure: 260909_plots/260917_level2_work_and_friction.pdf.

The measurement agrees, and the leading cost is second order. For $u \leq 0.05$ the fitted slope is -0.50 ± 0.67 , zero. Through the origin over all eight speeds, Au gives $\chi^2/\text{dof} = 9.2$, Au^2 gives 2.3, and on the three speeds carrying signal ($u \geq 0.1$) the quadratic form fits with $\chi^2/\text{dof} = 0.32$ and $A = 25.2$ while the cubic coefficient drifts $220 \rightarrow 124$ and is excluded. A matches $N_s m/2 = 25.0$: the excess work is the kinetic energy of the compression flow the piston leaves behind, and the gas indeed carries 65–70% of $N_s m u$ of momentum when the piston stops.

Level 1 re-passes on the correct form. Extrapolating with $a + Au^2$: $W(0) - W_{\text{qs}}^{\text{finite}} = -0.0151 \pm 0.0060 k_B T$, 1.7σ — on 760 trajectories instead of 30, and the unphysical slope $b = -4.2 \pm 1.7$ of the old straight-line fit was the quadratic curve forced through a line.

Audit (validation/level2_zeta_audit_20260917.py). The held wall is the piston itself, not a proxy: the labels WR ($t = 0\text{--}207.5\sigma$) and PR ($210.5\text{--}598.5\sigma$) never overlap, one object renamed at release. The running integral is converged over bin widths 1.0 down to 0.0625σ . And the explanation makes a prediction that holds on unused data: at the compressed end of the path, measured from the post-stop hold of the same runs, the cancellation moves to 18.0σ against the predicted $L_f/c_s = 15.9\sigma$ (start of path 21.5 against 19.8). The zero-lag value is the parameter-free Enskog piston friction $hn\sqrt{8mkT/\pi}g(\eta)$ to within 4% at the start and 9% at the compressed end.

Update 2026-09-18, on the corrected geometry ($L = 38.75$, gas $\Delta x = 3.93$, $W_{\text{qs}} = 7.4685 \pm 0.0131$). Three things now stand. (i) **The u^2 term is mostly protocol-launched:** ramping the piston over $2L/c_s$ instead of stepping it gives $A_{\text{ramp}} = 11.30 \pm 2.45$ against $A_{\text{step}} = 25.91 \pm 4.28$, 3.0σ apart, with A_{ramp} 1.2σ from the steady-flow value $N_s m/6 = 8.3$ and 5.6σ below $N_s m/2$. Because the piston's first 0.25σ is outside the gas, it reaches the gas at 35–50% of u , so 11.3 is an *upper* bound. (ii) **Paper 1 now supplies the friction:** the divider linewidth at $\eta = 0.112$ gives $\Gamma = 0.314 \pm 0.038$ against Enskog's 0.331 (within its 12% error), hence $\zeta_{\text{hyd}} = 0.081 \pm 0.010$ and a predicted linear term $0.32 \pm 0.04 u$ — consistent with the measured -0.50 ± 0.67 and still below the resolution of 760 trajectories. (iii) **Eq. (7)'s compression form is confirmed:** at $u = 10$ and gas $\Delta x = 7.08\sigma$, $\langle W \rangle = 1802 \pm 48$ against $2N_s m(\Delta x/L)u^2 = 1828$, a ratio of 0.986. Over $\Delta x = 0.75\text{--}7.08\sigma$ at $u = 10$ the ratio falls monotonically $1.08 \rightarrow 0.99$; the residual excess at small Δx is not explained.

Open: the cancellation is a property of a *closed* compartment. A longer or absorbing box, where the sound does not return coherently, should restore $W_{\text{diss}} = \zeta u \Delta x$. One geometry change, no new code. The fast end ($u = 3, 5, 10$) is not started.

How to run it yourself

Speed ladder (760 runs, the $A = 25.2$ measurement): `bash experiments.energy.transfer/_run_scripts/level2_speed_ladder_part1.sh` then `...part2.sh` (~ 25 s and ~ 5 min). **Ramp against step** (360 runs, ~ 40 s): `.../level2_ramp-vs-step-and-fast.sh`, which also produces the fast end; `.../level2_fast_1sigma.sh` for the travel- 1σ set and `.../level2_equilibrium_baseline.sh` for the decomposition's reference profile.

Analysis: `python3 hspist3/validation/paper2_level2_20260917.py` (work ladder and the running ζ), `level2.zeta.audit_20260918.py` (bin-width convergence, both ends of the path), `paper2_ramp_fast_20260918.py` (ramp vs step, decomposition, fast end), `level2_Au.figure_20260918.py` (the $A(u)$ figure).

The ramp needs the second binary — make release `TG=00ALLINONE_ramp` in `hspist3/`. One ramp trajectory, with the stop snapshot the decomposition uses:

```
HD_PISTON_EVENTS=/tmp/ev.csv HD_STOP_SNAPSHOT=/tmp/snap.csv ./00ALLINONE_ramp \
--mode=edmd --experiment=energy_transfer --headless --quiet --edmd-acc=0 \
--seed-drift-order=drift-first --energy-transfer-summary=/tmp/s.csv \
--energy-transfer-trace=/tmp/tr.csv --particles=100 --particles-boxes=50,50 \
--particle-radius=0.5 --l0=39.25 --height=10 --num-walls=1 --wall-positions=39.25 \
--wall-mass-factors=1000000000 --piston-right-protocol-mode=ramp --piston-ramp-time=40 \
--velocity-right-piston-step=0.05 --max-right-piston-travel=3.93 --auto-piston-step \
--wall-hold-steps=12000 --steps=17000 --fixed-dt=0.4 --energy-measurement \
--eff-output=wall-ke --kbt1 --seed=9200
```

Swap `ramp` for `step` (and drop `--piston-ramp-time`) for the control; raise `--velocity-right-piston-step` to 3, 5 or 10 and set `--max-right-piston-travel=0.75` for the fast end.

7.3 Fluctuation theorem (only after the two limits pass)

With canonical initial conditions — Maxwellian velocities *not* rescaled (a seeder flag we do not have yet; drift-first is microcanonical in KE) —

$$\langle e^{-\beta W} \rangle = e^{-\beta \Delta F}, \quad \Delta F = N_s k_B T \int_{\eta_i}^{\eta_f} \frac{Z}{\eta} d\eta \quad (\text{isothermal; ideal: } N_s k_B T \ln \frac{A_i}{A_f}). \quad (8)$$

Note $\Delta F \neq W_{\text{qs}}$ of (4): adiabatic quasi-static work heats the gas, the isothermal free energy does not. The estimator is dominated by rare low- W runs; you need of order $e^{\beta W_{\text{diss}}}$ seeds, so restrict to $\beta W_{\text{diss}} \leq 3$ ($u \leq 0.02$ here) and 10^3 seeds.

Pass criterion

$-k_B T \ln \langle e^{-\beta W} \rangle$ equals ΔF within its bootstrap error at $u = 0.005$ and 0.01 ; report the bias at $u = 0.02$.

8 Level 3 — geometry C: one gas drives a spring-loaded wall

Why this step exists

First appearance of ordered energy. The observable of the whole paper, $\epsilon = E_{\text{spring,max}}/W_{\text{in}}$, is defined here in the simplest case where it has a prediction.

The free wall of mass M sits on a spring k and is pushed by the gas pressure. Linearise about equilibrium: the gas acts as a second spring of stiffness $k_{\text{gas}} = \partial(PH)/\partial x = N_s k_B T [Z + \eta Z']/L_0^2$ (adiabatic: $Z + \eta Z' + Z^2$ for fast motion), so the wall is an oscillator with

$$\omega_w^2 = \frac{k + k_{\text{gas}}}{M}, \quad T_w = \frac{2\pi}{\omega_w}. \quad (9)$$

A piston push of duration τ_{push} raises the gas pressure by ΔP . Two limits with closed forms:

$$\text{quasi-static } (\tau_{\text{push}} \gg T_w): \quad x_{\text{eq}} = \frac{\Delta P H}{k + k_{\text{gas}}}, \quad E_{\text{spring}} = \frac{1}{2} k x_{\text{eq}}^2, \text{ no overshoot}; \quad (10)$$

$$\text{impulsive } (\tau_{\text{push}} \ll T_w): \quad \Delta p_w = \int \Delta P H dt, \quad E_{\text{spring,max}} = \frac{k}{k + k_{\text{gas}}} \cdot \frac{\Delta p_w^2}{2M}. \quad (11)$$

In between, $\epsilon(\tau_{\text{push}}/T_w)$ interpolates; that curve is the first result figure of Paper 2.

8.1 Why the 2026-09-19 pilot could not have passed

Quasi-statically the gas and the spring are two springs in *series*. A piston compression Δx divides between them and the spring takes

$$x_{\text{eq}} = \Delta x \frac{k_{\text{gas}}}{k + k_{\text{gas}}}, \quad E_{\text{qs}} = \frac{1}{2} k x_{\text{eq}}^2 = \frac{1}{2} k \Delta x^2 \frac{k_{\text{gas}}^2}{(k + k_{\text{gas}})^2}. \quad (12)$$

This is *maximal* at $k = k_{\text{gas}}$ — a stiffer spring barely moves, a softer one stores nothing — and the maximum is

$$E_{\text{max}} = \frac{k_{\text{gas}} \Delta x^2}{8} = \frac{N k_{\text{B}} T f^2 (Z + \eta Z')}{8}, \quad f = \frac{\Delta x}{L}. \quad (13)$$

The capture is **second order in the compression fraction** f while W_{qs} is first order. That is the whole story: at $f = 0.1$, $N_s = 50$, $\eta = 0.10$, Eq. (13) gives $E_{\text{max}} = 0.098 k_{\text{B}} T$, and the wall carries $k_{\text{B}} T/2$ of thermal energy in that same coordinate. The pilot ran $k = 5$, a factor 99 stiffer than $k_{\text{gas}} = 0.0507$, so its prediction was $E_{\text{qs}} = 0.0039 k_{\text{B}} T$ — $128\times$ below the pedestal. *No choice of k rescues it*: the ceiling over all k is still $5\times$ below. The pilot's measured $\epsilon = 0.487$ was the pedestal, not a capture. Its numbers stand as the design finding.

Two ways out, and Eq. (13) says which is cheap. Raising f works but only as f^2 : $E_{\text{max}} = k_{\text{B}} T/2$ needs $f \approx 0.16$ at $N = 100$ or $f \approx 0.07$ at $N = 500$. The cheaper route is to stop asking the spring to catch the *quasi-static* part at all.

8.2 Where Level 3 actually lives: the acoustic pulse

Level 2 measured what a stepped piston launches *beyond* W_{qs} : Au^2 with $A_{\text{step}} = 25.9 \pm 4.3$. That excess is coherent acoustic energy — precisely the “ordered energy before it thermalises” this ladder exists to chase — and it clears the $k_{\text{B}} T/2$ pedestal at $u \geq 0.2$ ($Au^2 = 1.04, 6.5, 25.9 k_{\text{B}} T$ at $u = 0.2, 0.5, 1.0$). The wall is then a mass-loaded acoustic termination and the share a pulse can deposit is

$$\mathcal{T}_{\text{imp}} = \frac{4Z_w Z_g}{(Z_w + Z_g)^2}, \quad Z_g = \rho c_s H, \quad Z_w = M_s \omega_w = \sqrt{M_s(k + k_{\text{gas}})}, \quad (14)$$

maximal at $Z_w = Z_g$.

A trap this calculation exposed. In geometry C of the master box ($N = 100$ over $L = 78.5 \sigma$) $k_{\text{gas}} = 0.0246$, *half* the pilot's 0.0507 — doubling both N and L halves $k_{\text{gas}} \propto N/L^2$. And with $M_s = 200$, Eq. (14) is already matched at $k \approx 0$: $Z_w/Z_g = \sqrt{M_s(Z + \eta Z')/N}/c_s = 0.998$. So at $M_s = 200$ the impedance optimum and the quasi-static optimum *coincide*, both at the soft end, and a k scan alone cannot tell the two predictions apart. It also means the soft end is unrunnable for a different reason: the wall's thermal excursion $\sqrt{k_{\text{B}} T/(k + k_{\text{gas}})}$ is 4.5σ at $k = k_{\text{gas}}$, comparable to the 10σ spring compartment.

The discriminating knob is therefore M_s , not k : Eq. (12) does not contain M_s at all, so a scan at fixed k leaves the quasi-static prediction flat while Eq. (14) predicts an interior maximum. At $k = 0.5$ the match sits at $M_s \approx 9.4$.

What we run (2026-09-20)

Master box, geometry C (divider removed, one gas of 100 disks, $\eta = 0.1001$, $\Delta x = 7.96 \sigma = 10\%$), step protocol, 40 seeds per cell, event log on, trace 200 σ -time past the stop. **(a) k ladder:** $k \in \{0.05, 0.5, 5\} \times u \in \{0.1, 0.2, 0.5, 1.0\}$. **(b) M_s ladder** at fixed $k = 0.5$: $M_s \in \{2, 10, 50, 1000\} \times u \in \{0.2, 0.5, 1.0\}$, which brackets the impedance match at $M_s \approx 9.4$ while E_{qs} stays fixed at $0.035 k_B T$. Primary observable is the **ensemble-mean** $E_{\text{spring}}(t)$; the per-seed maximum is the wrong estimator and the pilot showed why (it is $\sim 3\times$ the coherent peak, because it samples the tail of the pedestal).

Pass criterion (2026-09-20)

(i) The ledger (2) closes with the spring term to 10^{-8} . (ii) The coherent peak E_{coh} of the seed-averaged $E_{\text{spring}}(t)$ stands more than 3σ above the pre-push pedestal in every cell with $u \geq 0.2$. (iii) $\epsilon_{\text{coh}} = E_{\text{coh}} / \langle W_{\text{in}} \rangle$ has an interior maximum in the M_s ladder at the M_s nearest the impedance match, which Eq. (12) cannot produce. Report ring-down time and the final energy split as before.

9 Level 4 — geometry B: transmission through a free divider

Why this step exists

Before stacking walls we need the transmission of one wall between two gases, and we already own half of this result: the slow mode of Paper 1 *is* the heat conduction of the divider.

Release the divider between gas 1 (pushed) and gas 2. Two channels carry energy across: the fast one, pressure waves moving the divider ($\tau \sim \tau_{\text{sound}}$), and the slow one, heat conduction by divider collisions. Paper 1 measured the slow one: correlation time $\tau_{\text{heat}} = 54\text{--}100$ acoustic periods at $M = 100\text{--}1000$, amplitude set by the isothermal stiffness. The adiabatic-piston literature predicts $\tau_{\text{heat}} \propto M/m$; we can now test that with the KE_L, KE_R columns, and it becomes a figure for free. The transmission observable is the energy that has crossed after the push has relaxed:

$$\mathcal{T}(t) = \frac{\Delta KE_{\text{gas } 2}(t)}{W_{\text{in}}}, \quad \mathcal{T}_{\text{fast}} = \mathcal{T}(t \approx 5\tau_{\text{sound}}), \quad \mathcal{T}_{\text{slow}} = \mathcal{T}(t \rightarrow \infty) \rightarrow \frac{1}{2} \text{ (equipartition of the excess heat).} \quad (15)$$

The prediction to test for the fast channel is the two-body energy-transfer fraction with the gas column as a mass:

$$f(m_1, m_2) = \frac{4m_1 m_2}{(m_1 + m_2)^2}, \quad m_1 \rightarrow \text{effective gas mass } N_s m, \quad m_2 \rightarrow M, \quad (16)$$

which is heuristic and is exactly what Level 4 measures the validity of.

What we run

Geometry B, $L_0 = 20$, $N_s = 50$ each side, $\eta = 0.196$, held wall at L_0 (mass 10^9), free divider at $1.5L_0$ with $M = 20, 50, 100, 200, 500, 1000, 2000$; piston step $u/c_s = 0.1$, travel 2σ ; 25 seeds. Run to $t = 200 \tau_{\text{sound}}$ so $\mathcal{T}_{\text{slow}}$ is reached.

Pass criterion

$\mathcal{T}_{\text{slow}} \rightarrow 0.5$ within 2σ for every M (a ledger-level check). $\tau_{\text{heat}}(M)$ from the exponential approach; report its exponent in M (expected $+1$). $\mathcal{T}_{\text{fast}}(M)$: compare with (16); the pass is not agreement but a *measured* curve with error bars and the location of its maximum in $\alpha = M/N_s$.

What we measured (2026-09-17)

Level 4: the slow channel is already measured; the fast channel is not started. From Paper 1, with the KE_L, KE_R columns: the slow divider mode is stationary (its variance is the same in both halves of a 1000-period record, ratio 0.72–1.25), survives a $100\times$ longer hold, and has correlation time 53–102 acoustic periods at $M = 100$ –1000. Its amplitude is the *isothermal* equipartition value $\langle x^2 \rangle = L^2/(2N_s[Z + \eta Z'])$ to 4 % for light dividers, while the adiabatic form overshoots by 37–62 %. It follows $T_L - T_R$ directly: correlation +0.978 ($M = 100$) and +0.910 ($M = 1000$) with the slow part of the compartment-temperature difference, positive sign (a hotter left compartment pushes the divider to $+x$), lag ≤ 0.25 periods at $M = 100$, and the pressure balance $x = (L_0/2)(\Delta T/T) Z/(Z + \eta Z')$ reproduces the amplitude to 6–7 %. Across the A2 size ladder its amplitude falls as $N^{-0.39}$ to $N^{-0.54}$ in units of the box, i.e. it is a finite-size fluctuation that vanishes as $N \rightarrow \infty$. What is *not* yet measured: $\tau_{\text{heat}}(M) \propto M$ over a mass ladder, and the fast channel $\mathcal{T}_{\text{fast}}$ — both need the geometry-B runs of this level.

How to run it yourself

Data: Paper 1’s A1 v2 campaign, already on disk — this level needs no new runs yet. **Analysis:** `python3 hspist3/validation/paper1.linewidth_20260918.py` gives $\Gamma(\eta)$ from the divider linewidth; `damping_test_20260915.py` and `A2_slow_mode_20260915.py` give the slow-mode amplitude and its finite-size scaling.

10 Level 5 — geometry D and more walls: the efficiency experiment

Why this step exists

This is the paper’s question: how much of the injected work arrives as ordered energy, and what the protocol, the masses and the number of walls do to it.

Observable, protocol-averaged over seeds:

$$\epsilon = \frac{E_{\text{spring,max}}}{W_{\text{in}}}, \quad 1 - \epsilon = \frac{\text{heat left in gases} + KE_{\text{walls at the peak}}}{W_{\text{in}}}. \quad (17)$$

Knobs and the prediction for each: piston speed u/c_s (shock heating above ≈ 0.3 ; Level 2 gives the friction), spring-wall mass and divider masses (impedance matching, Level 4), number of intermediate walls n_w (the collision-chain estimate: two masses $m_1 \rightarrow m_2$ transfer f directly; with one intermediate $\sqrt{m_1 m_2}$ the product of the two steps is larger, 0.36 vs 0.18 for $50 \rightarrow 1000$; with geometric steps of ratio 2.7, 0.45). The prediction is therefore *ordered*: geometric intermediate masses raise ϵ , arbitrary ones lower it.

What we run

Geometry D: $L_0 = 20$, $H = 10$, $\eta = 0.196$ in each gas compartment, spring wall $M_s = 1000$, k as in Level 3. Series 1: no intermediate wall, $u/c_s = 0.02, 0.05, 0.1, 0.2, 0.5$. Series 2: one divider, $M_d = 50, 100, 224, 500, 1000$ at $u/c_s = 0.1$. Series 3: $n_w = 0, 1, 2$ with geometric masses, and $n_w = 2$ with two “wrong” masses (50, 50 and 1000, 1000), at $u/c_s = 0.1$. 25 seeds each. Every series carries the ledger.

Pass criterion

Series 1: ϵ decreases monotonically with u/c_s and the loss $1 - \epsilon$ at small u is consistent with ζ from Level 2. Series 2: $\epsilon(M_d)$ has a maximum; report its position against (16). Series 3: $\epsilon(n_w = 2, \text{geometric}) > \epsilon(n_w = 1) > \epsilon(n_w = 0)$ and both wrong-mass chains below $n_w = 0$, each ordering at $> 2\sigma$. This ordering, with error bars, is the headline figure.

11 Level 6 — information (only after 0–5, and only with a stochastic protocol)

With a protocol x_t drawn from a Markov chain and a coarse-grained system state s_t , Still, Sivak, Bell and Crooks give

$$\beta \langle W_{\text{diss}}[x_t \rightarrow x_{t+1}] \rangle \geq I_{\text{mem}} - I_{\text{pred}} = I_{\text{nost}}, \quad (18)$$

left side from the ledger (Levels 0–2), right side from joint histograms over 10^3 – 10^4 trajectories. Deterministic protocols have nothing to compute here, which is why Levels 0–5 use them and Level 6 cannot. This level is a separate paper’s worth; it is listed so the logging of Levels 0–5 is designed to serve it (per-interval work, x_t, s_t).

12 Order, cost, and the two flags we need first

Level	Runs	Needs	Status
0	0 (existing data)	nothing — the log exists	passed: 8×10^{-12} (energy), 1×10^{-13} (momentum)
1	1430 done	finite-box Z along the path	passed: $-0.015 \pm 0.006 k_B T$, 1.7σ
2 (slow end)	760 done	nothing new (ζ from existing holds)	passed, prediction corrected: no linear term; $W - W_c$
2 (fast end)	300	nothing new	not started
2 (Jarzynski)	2×10^3	canonical seeder flag	not started
3	125	spring + KE logging (exists)	not started
4	175	nothing new	slow-mode data already partly exists
5	375	multi-wall driver	not started

Two code items, both logging or seeding, no physics: the per-event wall-impulse log (serves Level 0 momentum *and* Level 2 ζ) and the canonical seeder (Jarzynski). Level 5 additionally needs the multi-wall driver. Everything else runs on the binary as it is. Total is about 3000 trajectories at $L_0 \leq 40$, i.e. hours, not days — the expensive part of Paper 2 is the analysis discipline, not the machine.

The one sentence per level, for the paper

0. Energy and momentum are conserved to round-off in every accepted trajectory.
1. The slow-push limit of the work equals the quasi-static work of the finite box.
2. In a closed box the friction coefficient measured from equilibrium fluctuations integrates to zero at the sound-crossing time, so the excess work is second order in the piston speed and equals the kinetic energy of the compression flow; at high speed it is single-collision work with no free parameter.
3. A spring-loaded wall driven by one gas stores the predicted fraction of the work in its two limits.
4. A free divider transmits energy through a fast pressure channel and a slow heat channel, the latter being the mode found in Paper 1.
5. The fraction of work delivered as ordered energy depends on speed, mass and the number of walls in the predicted order.